

Higher Tier only questions

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		$\text{Na}_2\text{S}_2\text{O}_3$ accept atoms in any order		1		1	1	
		(ii)		solution in water / dissolved in water	1			1		
		(iii)		sulfur forms as a solid / sulfur is insoluble / sulfur is precipitated (1) solid blocks view of cross (1)	1	1		2		2
	(b)	(i)		the mean of the results collected at 45°C / 24.3 (1) either of following for (1) - the middle value is very different to the other two / is an anomaly - the middle value should not have been used in calculating the mean			2	2		
		(ii)		the higher temperature the faster the reaction / higher the rate (1) particles have more energy / move faster (1) any of following for (1) - frequency of collisions increases - more collisions per unit time - more collisions have minimum required energy - more collisions achieve activation energy - more collisions are successful	2		1	3	1	
				Question 4 total	4	2	3	9	2	2